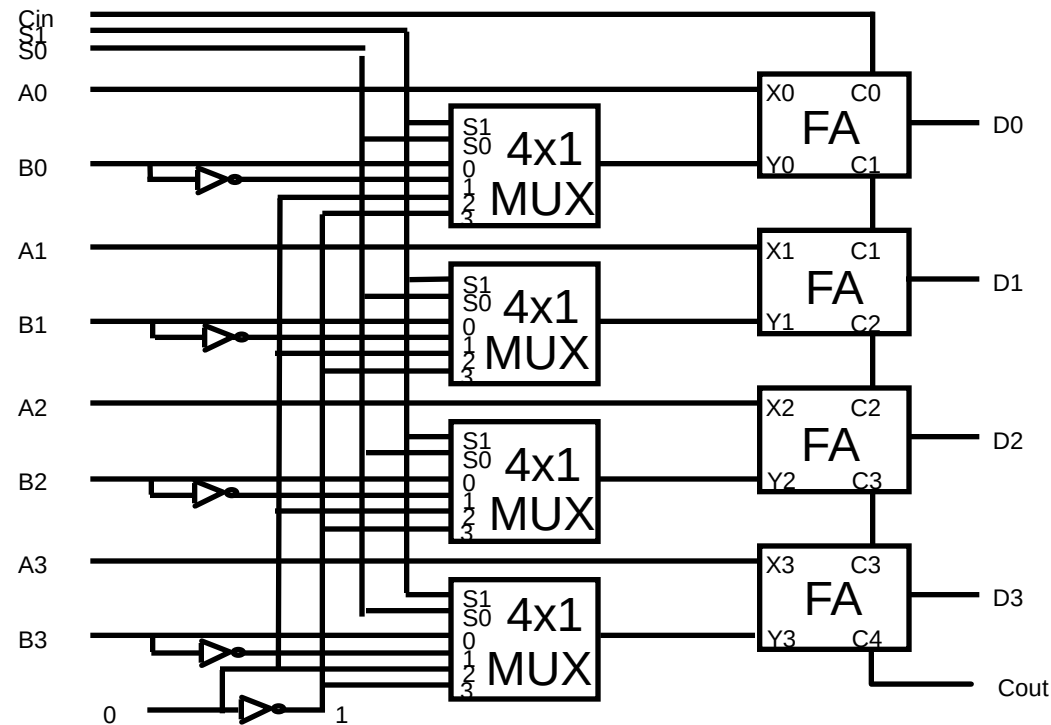


ARITHMETIC CIRCUIT



S1	S0	Cin	Y	Output	Microoperation
0	0	0	B	$D = A + B$	Add
0	0	1	B	$D = A + B + 1$	Add with carry
0	1	0	B'	$D = A + B'$	Subtract with borrow
0	1	1	B'	$D = A + B' + 1$	Subtract
1	0	0	0	$D = A$	Transfer A
1	0	1	0	$D = A + 1$	Increment A
1	1	0	1	$D = A - 1$	Decrement A
1	1	1	1	$D = A$	Transfer A



Arithmetic Circuit

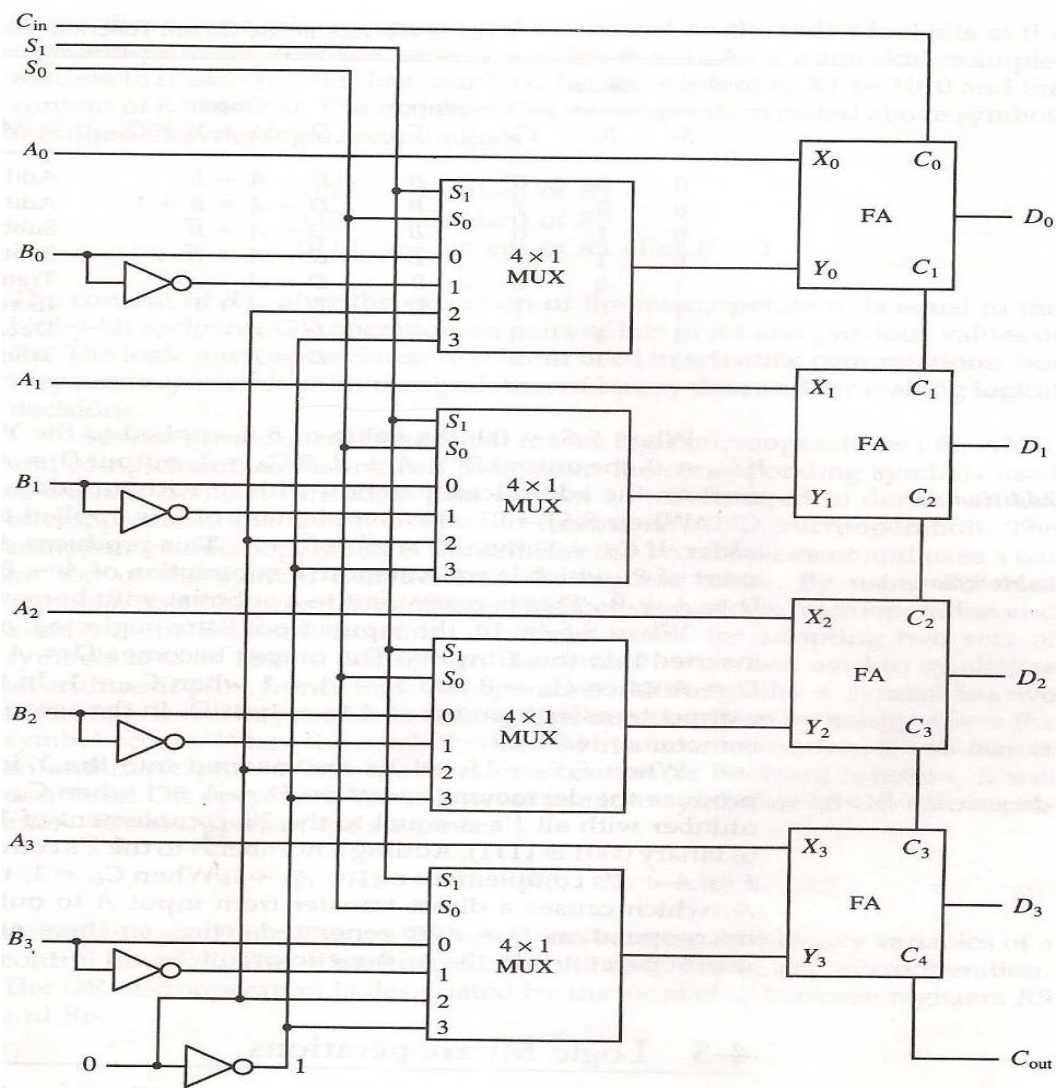


TABLE 4-4 Arithmetic Circuit Function Table					
Select			Input	Output	Microoperation
S_1	S_0	C_{in}	Y	$D = A + Y + C_{in}$	
0	0	0	B	$D = A + B$	Add
0	0	1	B	$D = A + B + 1$	Add with carry
0	1	0	$\overline{B}$	$D = A + \overline{B}$	Subtract with borrow
0	1	1	$\overline{B}$	$D = A + \overline{B} + 1$	Subtract
1	0	0	0	$D = A$	Transfer A
1	0	1	0	$D = A + 1$	Increment A
1	1	0	1	$D = A - 1$	Decrement A
1	1	1	1	$D = A$	Transfer A



LOGIC MICROOPERATIONS

- Specify binary operations on the strings of bits in registers
 - Logic microoperations are bit-wise operations, i.e., they work on the individual bits of data
 - useful for bit manipulations on binary data
 - useful for making logical decisions based on the bit value
- There are, in principle, 16 different logic functions that can be defined over two binary input variables

A	B	F_0	F_1	F_2	...	F_{13}	F_{14}	F_{15}
0	0	0	0	0	...	1	1	1
0	1	0	0	0	...	1	1	1
1	0	0	0	1	...	0	1	1
1	1	0	1	0	...	1	0	1

- However, most systems only implement four of these
 - AND ($\wedge$), OR ($\vee$), XOR ($\oplus$), Complement/NOT
- The others can be created from combination of these



LIST OF LOGIC MICROOPERATIONS

- List of Logic Microoperations
 - 16 different logic operations with 2 binary vars
 - n binary vars $\rightarrow 2^{2^n}$ functions
- Truth tables for 16 functions of 2 variables and the corresponding 16 logic micro-operations

<i>Boolean Function</i>	<i>Micro-Operations</i>	<i>Name</i>
$F0 = 0$	$F \leftarrow 0$	Clear
$F1 = xy$	$F \leftarrow A \wedge B$	AND
$F2 = xy'$	$F \leftarrow A \wedge B'$	
$F3 = x$	$F \leftarrow A$	Transfer A
$F4 = x'y$	$F \leftarrow A' \wedge B$	
$F5 = y$	$F \leftarrow B$	Transfer B
$F6 = x \oplus y$	$F \leftarrow A \oplus B$	Exclusive-OR
$F7 = x + y$	$F \leftarrow A \vee B$	OR
$F8 = (x + y)'$	$F \leftarrow (A \vee B)'$	NOR
$F9 = (x \oplus y)'$	$F \leftarrow (A \oplus B)'$	Exclusive-NOR
$F10 = y'$	$F \leftarrow B'$	Complement B
$F11 = x + y'$	$F \leftarrow A \vee B$	
$F12 = x'$	$F \leftarrow A'$	Complement A
$F13 = x' + y$	$F \leftarrow A' \vee B$	
$F14 = (xy)'$	$F \leftarrow (A \wedge B)'$	NAND
$F15 = 1$	$F \leftarrow \text{all 1's}$	Set to all 1's



Logic Microoperations

Boolean function	Microoperation	Name
$F_0 = 0$	$F \leftarrow 0$	Clear
$F_1 = xy$	$F \leftarrow A \wedge B$	AND
$F_2 = xy'$	$F \leftarrow A \wedge \bar{B}$	
$F_3 = x$	$F \leftarrow A$	Transfer A
$F_4 = x'y$	$F \leftarrow \bar{A} \wedge B$	
$F_5 = y$	$F \leftarrow B$	Transfer B
$F_6 = x \oplus y$	$F \leftarrow A \oplus B$	Exclusive-OR
$F_7 = x + y$	$F \leftarrow A \vee B$	OR
$F_8 = (x + y)'$	$F \leftarrow \overline{A \vee B}$	NOR
$F_9 = (x \oplus y)'$	$F \leftarrow \overline{A \oplus B}$	Exclusive-NOR
$F_{10} = y'$	$F \leftarrow \bar{B}$	Complement B
$F_{11} = x + y'$	$F \leftarrow A \vee \bar{B}$	
$F_{12} = x'$	$F \leftarrow \bar{A}$	Complement A
$F_{13} = x' + y$	$F \leftarrow \bar{A} \vee B$	
$F_{14} = (xy)'$	$F \leftarrow \overline{A \wedge B}$	NAND
$F_{15} = 1$	$F \leftarrow \text{all 1's}$	Set to all 1's

TABLE 4.5 Truth Tables for 16 Functions of Two Variables

x	y	F_0	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8	F_9	F_{10}	F_{11}	F_{12}	F_{13}	F_{14}	F_{15}
0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
0	1	0	0	0	0	1	1	1	1	0	0	0	0	1	1	1	1
1	0	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1
1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1

